**Wang Zining**

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## Education

**Hong Kong University of Science and Technology (HKUST)** — PhD Program
Division of Public Policy — 10/2025 - Present
Investigating the intersection of Deep Learning, LLMs, and 3D Visualization to tackle complex urban planning challenges and facilitate Evidence-based Planning.

**Kyushu University** — Master degree
GPA 3.53/4 — 10/2023 - 08/2025
International Program in Public Economics
Relevant Coursework: Econometrics 4.0, Economic Theory 3.0, Information Analysis 4.0, Asian Economic Research 4.0
Recipient of the Monbukagakusho Honors Scholarship (JASSO)

**Jiangsu Ocean University** — Bachelor degree
GPA 4.53/5 ( Ranked Top 5 in the class) — 08/2019 – 06/2023
Major: International Economics
Relevant Coursework: Macroeconomics 5.0, Microeconomics 4.0, Econometrics 5.0, Finance 5.0, Linear Algebra 4.0, Statistics 3.5
Recipient of University-Level Outstanding Thesis Award
Recipient of First-Class Scholarship for Outstanding Academic Achievement

## Publications

*Journal metrics in parentheses are from Journal Citation Reports (JCR) 2025.*

**Wang Zining**, Pengyu Zhu. (2026) Discovering the Temporal Evolution of Spatial Spillovers of Green Innovation in China: An Innovative Approach Using Dynamic Spatial-Temporal Weight Matrices. *Annals of the American Association of Geographers*, 1–27. (JIF: 3.8; Rank: 23/174 in Geography, Q1, 13%, founded in 1911)

Zhu Pengyu, **Zining Wang (co-first author)**, Xuan Zhang, Kailai Wang. (2026) Public Transit Funding and Transit Use: A Panel Study of U.S. Urbanized Areas 2010–2019. *Transportation Research Part A: Policy and Practice*, **212**, 105133. (JIF: 7.2; Rank: 21/626 in Economics, Q1, 4%)

Yu Tao, Pengyu Zhu, Dehao Shi, **Zining Wang**. (2026) State Entrepreneurialism under Economic Transition: Bounded Rationality in China's High-Speed Rail New Town Governance. *Urban Studies*, online first. (JIF: 4.4; Rank: 11/86 in Urban Studies, Q1, 12%, founded in 1964)

**Wang Zining**, Pengyu Zhu, Bing He. (2026) How does China's Low-Carbon City Pilot Program Promote Green Investment: A Spatial–Temporal Analysis at the City and Firm Levels. *Humanities and Social Sciences Communications*, **13**, 1416. (JIF: 4.8; Rank: 9/275 in Social Sciences, Interdisciplinary, Q1, 3%)

Zhu Pengyu, **Zining Wang**. (2026) Is Taxi a Net Substitute or Complement for Public Transit? Evidence from High-Granularity Taxi GPS Data. *Humanities and Social Sciences Communications*, online first, 10.1057/s41599-026-08517-w. (JIF: 4.8; Rank: 9/275 in Social Sciences, Interdisciplinary, Q1, 3%)

**Wang Zining**, Pengyu Zhu. (2026) Spatiotemporal Relationship Between Green Product Trade and Air Pollution in China. *Journal of Urban Management*. **Accepted** (JIF: 5.6; Rank: 10/86 in Urban Studies, Q1, 11%)

Zhu Pengyu, **Zining Wang**, Renu Singh, Xinying Tan. (2026) China's Model of Technology Leapfrog: A Case Study of Electric Vehicle Policies and the Development of Green Technology. *Renewable and Sustainable Energy Reviews*, **226**, Part D: 116414. (JIF: 18.0; Rank: 4/114 in Green & Sustainable Science & Technology, Q1, 3%)

**Wang Zining**, Pengyu Zhu. (2025) Trade of Eco-Friendly Products and $CO_2$ Emissions in China: A Spatial Spillover Analysis within the EKC Framework. *Journal of Cleaner Production*, **516**, 145734. (JIF: 10.7; Rank: 27/395 in Environmental Sciences, Q1, 7%)

Zhu Pengyu, Jianqi Li, **Zining Wang**. (2025) Identifying Functional Agglomerations and Urban Centers Using Open-Source Data and Machine Learning: Framework, Applications and Planning Implications. *Journal of Urban Management*, **14** (4), 1099–1114, 10.1016/j.jum.2025.04.004. (JIF: 5.6; Rank: 10/86 in Urban Studies, Q1, 11%)

**Wang Zining**, Bing He, Toshiyuki Fujita. (2025) Does Environmental Regulation Affect Inward Foreign Direct Investment? Evidence from China. *Applied Economics Letters*, **32** (19), 2877–2883. (Founded in 1994)

**Wang Zining**, Hao Wang, Bing He. (2023) The Investment Development Path Theory: Evidence from China. *Asian Journal of Economics and Business*, **4** (2), 179–191.

**Wang Zining**, Bing He. (2022) Does Trade Openness Influence the Development of Two-way Direct Investment? Evidence from Jiangsu Province, China. *Modern Economy Management*, **1** (1), 6.

## Papers Under Review in Peer-reviewed Journals

**Wang Zining**, Pengyu Zhu. Does Green Space Always Pay? Spatial Thresholds, Playability, and Scale-Dependent Capitalization in China's Housing Market. *Landscape and Urban Planning*. **Submitted** 30 March 2026

**Wang Zining**, Teklebirhan Alemnew Gebeyehu, Pengyu Zhu. Green Scarcity and Housing Stratification: Who Captures the Value of Urban Green Space in a Hyper-Dense City? *Land Use Policy*. **Submitted** 1 May 2026

**Wang Zining**, Pengyu Zhu, Xiaodong Chen. Health Returns to Water Infrastructure: Evidence from a Policy Discontinuity in China. *Nature Health*. **Submitted** 12 March 2026

**Wang Zining**, Pengyu Zhu. The 15-Minute Green Necklace: How Granular Parks Enhance Land Value. *npj Urban Sustainability*. **Submitted** 5 April 2026

Zhu Pengyu, **Zining Wang**. Do Local Land Use Regulations Combine to Reduce Commute Duration and Improve Job Accessibility in U.S. Metropolitan Areas? *Cities*. **Submitted** 12 April 2026

Zhu Pengyu, **Zining Wang**, Jia Kang, Yuqing Guo. Non-linear Spatiotemporal Impacts of High-Speed Railway on Land Price. *Transportation Research Part A: Policy and Practice*. **Submitted** 1 February 2026

## Working Paper

**Wang Zining**, Pengyu Zhu. Generalized spatio-temporally weighted panel data model

> This paper proposes a **generalized spatio-temporally weighted panel data model (GSTW-PDM)** that jointly addresses three limitations of conventional spatial panel specifications: static weight matrices, the exclusion of cross-period spillovers, and the endogeneity of economically constructed weights.

**Wang Zining**, Pengyu Zhu. Global Systematic Variations in Urban Streetscapes Revealed by the PROSPECT Index

> We introduce the **PROSPECT Index**—a composite measure of eight perceptual dimensions capturing openness, enclosure, greenery, sky exposure, and structural complexity—derived from nearly ten million street-view images

spanning 668 cities across six continents. Using state-of-the-art semantic segmentation (SegFormer-B5), monocular depth estimation (MiDaS v2.1), and visual geometry networks (InVPT), we reveal systematic, geographically structured variation in urban streetscapes that transcends individual city design choices.

**Wang Zining**, Pengyu Zhu. Building Without Residents: A Political Economy of Ghost Cities in China

We develop a game-theoretic model in which local officials overbuild to win GDP tournaments, generating a Nash equilibrium where vacancy costs are systematically shifted onto successors. To detect and classify ghost cities at scale, we construct a 35-year longitudinal panel combining nighttime-light anomaly scoring with multimodal satellite image classification. **Ghost cities are not market failures but equilibrium outcomes of deliberately designed institutions.**

**Wang Zining**, Pengyu Zhu. Seeing Slums: Multi-Source Detection of Informal Settlements Across the Global South

We build a transferable detection pipeline integrating building height, POI, OSM, and nighttime light with frontier vision-language models capable of zero-shot cross-regional generalisation across Sub-Saharan Africa, South Asia, and Latin America. The result is the first global panel of informal settlement intensity, enabling measurement where data have never existed.

## Conference Presentations

2026 "Spatial Panel Regression (SPR): Discovering the Temporal Evolution of Spatial Spillovers" ***Lesvos Island, Greece***

2025 "Is Green Product Trade a Pathway to Better Air Quality in China?" ***Oxford University***, Oxford, United Kingdom

2025 "New Spatial-Temporal Weight Matrices" ***20th Anniversary Conference of the Japan Association for Applied Economics***, June 7–8, Hyogo Prefectural University (Kobe Commerce Campus), Japan

2024 "The Impact of Environmental Regulation on Green Investment in China" ***Japan Association for Applied Economics***, November 9–10, Nishinomiya, Hyogo, Japan

## Skills

**Technical:** Proficient: Stata, ArcGIS, GeoDa; R, LaTeX, Python; MATLAB.
**Language:** TOEFL 105, The Japanese-Language Proficiency Test.